is generally conceded that the Stephenson motion gives the best steam distribution. When the Walschaert gear is once properly set up, the valves will remain square much longer than those of the Stephenson, and this is a strong point favoring the Walschaert motion, for if the valves of the Stephenson motion do not remain square their better steam distribution is only temporary and the economy of their setting is lost.

The Stephenson link is made to move through wide angles by the influence of two eccentrics. This produces a wedging effect on the link block, strains the gear and causes lost motion. The Walschaert link is only moved through comparatively small angles by the influence of one eccentric, and consequently there is less strain and wear on the parts. Again, the up-and-down motion of the engine on its springs affects the steam distribution of the Stephenson motion to some extent, while with the Walschaert gear this movement has no disturbing effect on the valve motion unless the connection of the eccentric rod to the link is placed at too high a point above the center line of the axles. While the proper location of this connection should be as near the center line of motion as possible, it is generally found necessary to locate it above this line in order to retain the required amount of link block travel and at the same time keep the throw of the eccentric crank within certain limits. Usually, however, the location of this point is not high enough to cause a noted irregularity in the valve's motion unless the track is very rough.

The usual form of Walschaert gear imparts its motion to the valve in a more direct manner than does the Stephenson gear, since there is no offset rocker arm to offset the motion and spring. However, some roads have introduced a direct acting offset rocker arm in connection with the gear and have thereby been able to use the same style of cylinder pattern, whether applying the Walschaert or Stephenson motions to their engines.

In addition to the fact that the Walschaert gear is placed on the outside of the engine, it differs chiefly from the Stephenson link motion in that it requires but one eccentric or its equivalent for each cylinder to insure the movement of the valve and the proper distribution of the steam for both forward and backward motions. There are two distinct movements which control the movement of the valve. One from a single eccentric (usually in the form of an eccentric crank) and another from the crosshead. The eccentric throw governs the valve travel and reversing operations, while the crosshead motion governs the lead by moving the valve from its central position far enough to overcome its steam lap and enough farther to give the amount of port opening desired for the lead.

Another point of difference between the two motions is that the Walschaert gear gives a constant lead at all points of cut off, while the Stephenson motion, on account of the influence of two eccentrics, produces an increase or decrease in lead as the link is hooked up, depending upon whether open or crossed eccentric rods are used, open rods are invariably used when applying this motion to locomotives, and the lead may vary in some cases from blind in full gear to $\frac{5}{8}$ in. open in midgear. Since lead is a positive detriment to an engine when starting a heavy train, the variable lead of the Stephenson motion in this case would have a practical advantage over the constant lead of the Walscheart motion. Theoretically, the constant lead of Walschaert gear is a distinct disadvantage when applied to locomotives which are to be run at any considerable range of speed, and probably this one point of difference has had a large influence in retarding the progress of the use of the Walschaert gear.

It is still a matter of argument among experts as to whether or not lead as a port opening is really a necessity, for, while many advocate lead, it is still considered by some to be of no advantage, and even to be a detriment. It is generally conceded, however, that when running at low speeds only a small lead is needed, while at high speeds more lead is required. While these conditions are produced by the Stephenson motion, it is admitted that this increase of lead often becomes excessive at early cut-offs. The admission of steam in this case may take place too early and have a tendency to counteract the work done by the steam on the driving side of the piston. The exhaust steam is also cut off earlier, giving more than the proper amount of compression, which is only meant to be great enough to cushion the advancing piston and arrest the motion of the reciprocating parts at the completion of the stroke. Consequently, when these conditions become excessive they are a detriment to the speed of the engine.

The proper amount of lead will vary for different classes of service, and the Walschaert valve gear must be designed to give suitable lead for its particular class of service. This should seldom exceed $\frac{1}{8}$ in., as the advantage received from an earlier admission of steam would probably be more than offset by the increased compression and preadmission. Since the lead given by the Walschaert motion is governed by the designer, it follows that no alteration should be made in any of its parts to bring about an increase or decrease in its lead. If desired, however, the lead may be increased at the expense of the steam lap by cutting off the edge of the valve or rings, which would, of course, bring about an earlier admission and allow less expansion of steam. On the other hand, lead might be decreased by using another valve with more steam lap. Lead may also be increased or decreased by changing the distances between the connecting points of the lap and lead lever, but this is not considered good practice. It should be remembered that the valve, strictly speaking, is not a part of the valve gear.

With the Walschaert gear the block moves up or down in the link when the engine is reversed, instead of the link moving up or down on the block, as is the case with the Stephenson motion. The link is made to swing on trunnions through the influence of the eccentric crank. The link block, which may be in either end of the link, in turn transmits this movement to the valve through the radius bar, the forward end of which is connected to the combination lever. This connection between the radius bar and combination lever is supported and guided by the valve-stem crosshead, and forms a fulcrum on which the combination lever is made to swing by the action of the

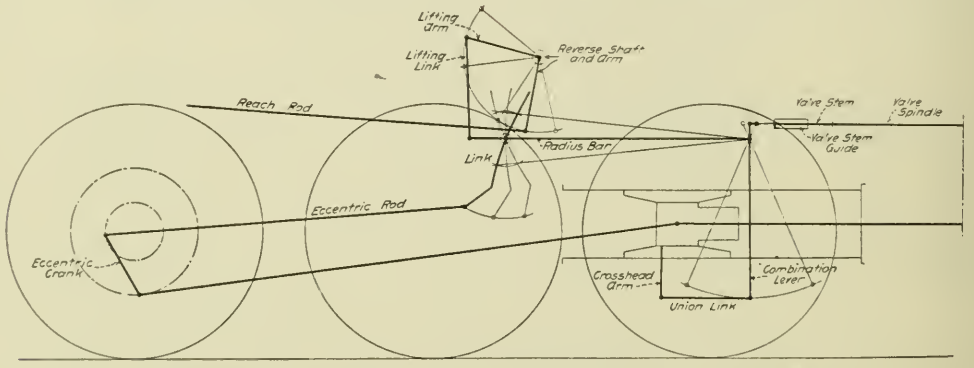


Diagram of Walschaert Valve Gear Applied to Mallet Locomotive.

piston crosshead. When outside admission valves are used, this connection or tulerum between the radius bar and combination lever is placed below the valve rod connection, while with inside admission valves this tulerum is placed above the valve rod connection. Thus it may be easily determined whether an engine with piston valves has outside or inside admission by noting where the radius bar connects to the combination lever.

The lower end of the combination lever is connected to an arm on the crosshead by means of the union link. The combination lever is commonly called the lap-and-lead lever, probably because its object is to move the valve from its mid position an amount equal to the lap and lead when the engine is on either dead center. The amount of movement that this lever gives to the valve for each stroke is therefore equal to twice the lap plus twice the lead. Since the combination lever receives a constant movement from the crosshead, its action must necessarily remain the same at all positions of the reverse lever, thereby explaining why the Walschaert gear gives a constant lead. The throw of the eccentric crank controls the mit al movement of the valve for both forward and backward motions.

The link governs the reversing operations, since it acts as a double rocker arm by controlling the movement of the link block in either the same or opposite direction to that of the eccentric rod, depending on the location of the block in the link. It follows, then, that, unlike the Stephenson motion, this gear may be classed either as a direct or indirect valve gear, depending upon whether the block is located above or below its fulcrum. As a rule, the Walschaert gear is designed so that the link block moves in the lower end of the link when the engine is running ahead. In this case the eccentric crank must always follow the crank pin by about 90 deg. when the engine has inside admission valves or precede the crank by 90 deg. when outside admission valves are used. Therefore, it depends on the position of the eccentric whether the engine will run forward or backward with the block in the lower end of the link.

The exact location of the eccentric crank should be at right angles with the plane of motion or 90 deg. to a line drawn from the point on the link at which the eccentric rod is attached through the center of the driving axle when the engine is on a dead center. Reversing the location of the blocks in the link for one motion makes it necessary to shift the eccentric crank 180 deg., thus determining whether the eccentric crank should follow or precede the main crank by approximately 90 deg.

Most articulated locomotives are arranged so that the link blocks for one engine are at the top of the link while the link blocks of the other are at the bottom of the link when the engine is in position to run. Since all the link blocks must be moved at the same time by the reverse lever, this arrangement was made in order to produce a balancing effect, and thus reduce the strain required at the reverse lever. In this case the eccentric crank of one engine follows while the eccentric crank of the other precedes the main crank, provided both engines use the same style of valves—both inside or both outside admission.

The throw of the eccentric crank is necessarily much greater than the throw of the ordinary Stephenson eccentric, since the Walschaert link acts as a reducing rocker arm and transmits to the valve only about one-third of the throw of the eccentric. The ratio between the link foot radius and the link block radius is about three to one when the block is in full gear position. As the reverse lever is hooked up, the link block is brought nearer to the center of the link and the horizontal movement of the block becomes less; this produces a shorter valve travel, and consequently an earlier cut-off. Therefore, when the block is brought in line with the center of the link it can give no movement to the valve. If in this case the engine were towed ahead, the valve would still receive a movement from the crosshead connection, and the amount of this movement, as explained before, would be equal to two times the lap plus the lead for each stroke of the crosshead. The fact that the crosshead movement controls this part of the valve's movement makes it easier to secure an equal cut-off at each end of the cylinder.

When an engine using the Walschaert gear is on a dead center the link block may be moved from one end of the link to the other without giving any movement to the valve. It is therefore necessary that the length of the radius bar should be equal to the radius of the link; in fact, it may be said that the length of the radius bar governs the radius of the link.

There are two general forms of lifting devices for the radius bar. In any case, the suspension point of the lifter must be so placed that the link block will travel as nearly as practicable on a chord of the arc described by any point of the link wherever the block happens to be when the link is swung into one of its extreme positions. The above conditions are most closely approximated by a lifter through which the radius bar slides, not swinging with the link. Practically the same results may be obtained by a properly suspended hanger, although the link block slip will be more in the backward than in the forward motion. In setting up this motion it must be remembered that the positions of the link blocks must be the same on both sides of the engine, regardless of the position of reverse lever.

Unlike the Stephenson gear, the Walschaert motion will not admit of experimental changes or readjustments in the roundhouse or on the road. There is no part that can be lengthened or shortened outside of the general repair shop, and even then there should be no necessity for alterations in the motion work. The proportions of the various parts of the Walschaert gear cannot be determined experimentally, nor should any change in setting the valves be made unless the effect of the change is known in advance. In setting up this valve motion much more dependence is placed upon the accuracy of the machining operations and in the checking up of the frame and cylinder measurements than is permitted by the Stephenson motion, which allows itself to be adjusted to compensate for certain irregularities in the machining or erecting of the different parts. Therefore, assuming that all work has been properly done, it is a much simpler matter to set the valves of the Walschaert gear than those of the Stephenson. It is also much easier and more convenient to repair the Walschaert than the Stephenson motion.

It is claimed by some that the Walschaert gear has some objectionable features which are overcome by some of the new pin-connected outside gears which use no link. Again, it is claimed that when rounding curves the Walschaert gear is subject to disturbances that tend to produce irregularity in the valve motion. However, it is well to remember that the Walschaert gear not only meets the requirements of modern locomotive design, but can be as easily applied to locomotives of odd design, in which case the Stephenson motion would probably require the addition of complicated parts, which would add considerable to its weight. On some types of modern locomotives, the Mallet compound for example, it is absolutely necessary to use an outside gear, owing to the space between frames being entirely taken up by the pipes and supports of the boiler.

A consular report says that the President of Brazil recently laid the corner stone of one of two 32,000-h.p. electric power stations which are to furnish the motive power for an electric railway between the port of Victoria and the city of Itabira, in the state of Minas Geraes. The railway will be about 400 miles long, and will tap one of the greatest sources of Brazil's wealth in iron ore. One of the large waterfalls on the river Doce will furnish the power for one of the stations, and a large waterfall on a tributary stream, power to the second station. At Itabira connection will be made with the Central of Brazil, with which it will have mutual traffic agreements. It is expected to thus land at least 3,000,000 tons of ore annually at Victoria at an estimated cost for transportation of about four mills per ton-mile. About 250 miles of an existing steam railway are to be utilized for this railway, the gage to be changed from narrow to standard. The electrical equipment is to be furnished by a British concern, the entire project being financed by British capital interested in the development of the iron-ore deposits.

paring the measurements when the boiler was hot, that the difference of expansion of both sheets was too significant to be ignored.

To compensate for the difference in the amount of sheet expansion, adjustments were made in the application of the flexible staybolts by turning back the bolt head from its sleeve seat certain parts of a turn, before riveting over the firebox end of bolt. This was done in the outer rows of the side sheets, back head and throat sheet, the amount of release being varied according to the size of the firebox and the relative showing in the difference of sheet expansion. By so doing, it enabled the firebox and outer shell to expand under less restraint, and the staybolts to assume the load with less liability of being strained than under former practice of no adjustment. The results obtained proved satisfactory, the experiment and practice extending over a sufficient number of years to obtain comparative data and form reasonable conclusions, not only as to the life of the staybolts, but as to the condition of seams, rivets, side and flue sheets, all of which showed no perceptible indication of strain or distortion.

Further investigation by others not connected with but interested in the aforementioned experiment, disclosed the fact that when flexible staybolts were adjusted, especially in the throat sheets, to compensate for the longitudinal extension of the wrapper sheet over the firebox sheets, in instances where this method was applied to old fireboxes, when the staybolts were removed for firebox renewals it was discovered, by nicking and bending the iron in the bolt proper, that the material was as fibrous and as strong as when first incorporated. The bolt head showed a distinct bearing surface at the sleeve seat, demonstrating that under the adjustments made, it served its purpose as a stay, resisting the load due to pressure without becoming strained, and that the difference in the relative sheet expansion is largely compensated for by the adjustment which the flexible stay affords. Expansion of all heat absorbing parts in a locomotive firebox, if restricted, will effect the entire assemblage and prove disastrous to the weaker part of the construction.

The value of the difference in the amount of expansion between the firebox sheets and the roof or wrapper sheet has never been seriously considered to any extent until quite recently. It was supposed that most of the expansion was confined to the firebox proper, while the relative amount of expansion between the wrapper and the firebox sheets, and the resultant effect therefrom on the complete construction was overlooked in most instances, as a vital point to consider in dealing with methods of staying.

The distortion of sheets has more or less lengthened the life of the rigid stay, while the sheets deteriorated by reason of continual bending. While the relative expansion of the outer shell and the firebox varies with each type of boiler, and is largely influenced by methods of firebox and boiler operation, the difference in the amount of expansion is at times so great as to throw an excess stress on staybolts in the throat, back head and rigid corners of the side sheets. Regardless of sheet distortion, the bolts then assume a load too severe to withstand and are gradually rendered weak, and break.

The flexible staybolt has had much to contend with on points of cost compared with the rigid stay, and not until the question of economy of maintenance was earnestly considered, where ultimate costs determined the true value of a product based on service rendered, could we point to any great advance in the use of the flexible stay, or the methods of staying. Flexible staybolts found but little favor until within the last six years. The early designs were either too large or ungainly, or too weak in sections to cope with firebox conditions. The two-piece bolt of the Johnston type made great headway as a simple and strong device to serve the purpose of a water space stay, but was soon declared obsolete by reason of the fact that inspection was impossible without removing the entire bolt. Incrustation in bad water sections rendered the connection rigid, cementing both the plug and the bolt intimately together.

The principle of the three-piece design for a flexible stay, was recognized as the most acceptable on which to base improvements and modifications soon led to shapes and sections involving the round head bolt. This affords greater shearing strength, and more readily releases itself under conditions of incrustation, than similar designs of the flat or sloping heads. The line of demarkation between serviceable and non-serviceable staybolts of the flexible type is determined largely on the merits which enable the bolt to operate under conditions of incrustation. The flaring mouth on the end of the sleeve of the round head bolt extending toward the water space serves to loosen and throw out all deposits which collect within the water space in adjusting itself to accommodate the relative sheet expansion under various temperatures.

The round head bolt design with its strong sections of bolt area, capped over with sufficient clearance between the cap and the bolt head to allow for suitable adjustment to the difference of sheet expansion, affording a ready means of inspection, with its several sizes of sleeves to suit the various angles and sheet contours of boiler design, adapted to take several diameters of bolt and affording interchangeability of parts, simple and economic in its application, forms an assemblage which readily establishes itself as an acceptable article of economic merit and serviceable value, and is fast proving the fact that flexible connections in firebox construction are absolutely needed.

The demand for flexible staybolts is due to the effort to curtail not only the resultant expense of staybolt renewals, but more largely to reduce the number of shoppings of engines for the purpose of renewing broken staybolts and repairing cracked fire sheets, which not only deprives the service of power, but contracts the earning value of each engine in direct proportion to the maintenance cost and the service rendered.

Essential to the economic solution of overcoming cost of maintenance in firebox repairs is the step that will first favor the construction of parts to more readily resist the stress of expansion without the strain of distortion, heretofore found so disastrous to all materials involved, and provide sufficient means to allow all heat absorbing surfaces to act under less restraint than that afforded by the complete system of rigid staying. Until locomotive boilers are measured to obtain the true amount of the difference of expansion from the cold to the hot state, and provisions are made to accommodate the relative amount of expansion between the wrapper sheet and firebox, the cause of staybolt breakage, fire sheet and flue sheet cracking, will remain an issue of misleading conclusions, if as in the past, quality of material forms the basis of solution, and the destructive forces due to the inequalities of expansion are ignored.

The flexible or adjustable staybolt so far has favored the theories of those who are mindful of the fact that steam boilers must breathe or expand under less restraint than that allowable with rigid staying. In the experiments and practical demonstrations made in the application and adjustment of the flexible staybolt to high pressure locomotives, covering, in many instances, years of constant service without staybolt breakage, side seams leaking, and sheets cracking, and obtaining a great reduction of breakages in most cases, the value of the service rendered compared to the cost of repairs is a unit of economic comparison, which not only credits the flexible staybolt as being superior to the rigid stay in localities of greatest sheet expansion, but also points conclusively to the advantage gained in the judicious use of the flexible or adjustable staybolt as a compensating medium of great utility in affording sufficient flexibility to the firebox construction, enabling such to more safely cope with the alternating stresses due to expansion.

The general recognition given to the flexible staybolt in late years, as an article of economic use and of practical utility, and to the methods of its application and adjustments necessary in the rigid localities of greatest sheet expansion to relieve the bolt of excess tension; the judicious layout in the effort to fully cover the breaking zone according to each type of firebox, and to minimize the effects of the alternating stresses so disastrous to staybolts—all features of vital importance—have to the extent of the consideration given and the practice followed, accomplished much to advance the general proposition of eliminating staybolt breakage and fire sheets cracking, inasmuch as the cause of breakage is traced to its true source.

President Gomez has signed a decree contracting with the Cuban Central Railways, Limited, for the construction of new railways and granting the subsidy prescribed by the law of July 5, 1906. The railway lines concerned are those from Sagua la Grande to Corralillo, by way of Rancho Veloz, and from Cifuentes to La Esperanza, by way of San Diego del Vallo. A bill was introduced in the Chamber of Representatives under date of November 16 to authorize the executive to contract for building a railway from Guantanamo to Baracoa, with application of the subsidy law.